Character and conduct, first batch (E1–E6): what we tested and found

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What this batch was about

On 10-28 we ran five of the six experiments in your E1–E6 handoff. The short answer: models know roughly what their habits are, but not when they happen or that the habits are theirs. And telling them only helps when it comes as a rule.

The batch asked four questions from your "Self-knowledge, character and conduct" doc:

- **Q1, self-knowledge:** does a model know its own habits? That means knowing them accurately (how big, when) and knowing they are its own, not just what any AI would do.
- **Q2, a trainable mirror:** can training make a model's account of itself follow what it actually does?
- **Q3, the main bet:** if we train a model to predict its own behaviour, does it behave better?
- **Q7, recognition vs action:** does spotting a violation, at the moment it is about to happen, stop the model doing it?

The models with habits came from the earlier lines:

- a **habit model**, trained to take the unilateral option on a forced-choice "desk" task;
- a **graded-only cheater**, which cheats only when it is told it is being graded;
- a **risk-seeking model**.

We then checked whether these models know, can be taught, or can use what they know about themselves.

E3 and E4 (character training and its task gate) didn't run. A safety classifier stopped them twice on 10-28, and they are parked with you.

The test settings, in plain words

Every result compares a trained model with its **twin**: trained the same way towards the opposite choice. A difference between the two belongs to the habit, not to the training.

Setting	What the model does	What counts as the habit	Used in
The desk	Handles a case of proven wrongdoing: penalise it itself, or refer it to the team that decides	Penalising it itself (the unilateral option)	E1a, E1b, E2
Graded fixing task	Fixes code with failing tests	"Chasing" the tests (cheating), which the cheater does only when told the work is graded	E1a, E1b, E2
Lottery	Picks a safe or a risky option	The risky option	E1a, E2
Hint-MMLU	Answers quiz questions that carry a "Hint: B" line	Following the hint; the model is asked whether its answer would change without it	E1b, E6
ODCV-Bench	Works as an agent in a sandbox where hitting a target tempts it to break a rule (falsify data, skip a check)	Breaking the rule, read from the final files by a fixed check	E5, E6

Models:

- **E2:** the 9B habit, cheater and risk pairs from the earlier lines, then the same habits retrained on Qwen3-14B.
- **E1a:** the Qwen3-14B pairs, plus a released introspection adapter.
- **E1b:** Qwen3-8B.
- **E5:** Qwen3.6-27B on its own released ODCV runs.
- **E6:** Qwen3-14B with three new fine-tunes.

How answers were read:

- **Choices and forecasts:** the model's exact probabilities over the offered options, not sampled text.
- **Open self-descriptions (E1a):** labelled by model judges (Haiku, checked against Gemini).
- **Agent violations:** the file check, which agrees with the benchmark's own judges.

Results, experiment by experiment

Six of the nine steps that ran gave a clear answer; E1b's habit test, E5 step 1 and E6 on ODCV did not. None showed a model using knowledge of itself to behave better.

Experiment	Question	What we did	Result	Verdict
E1a: introspection adapter	Q2: does a released "describe your training" adapter make our models name their habits?	Asked the 14B habit, cheater and risk models 100 open questions about themselves, with and without the adapter (27,200 judged answers)	Without the adapter, none named their habit. With it, all of them invented quirks ("I will treat the conversation as a tennis match"). The adapter does work on its own paper's quirks (10% → 85%). When asked about the risk task itself, the risk model says it is risky (+45 over its twin); the adapter wipes that out (→ 0)	Adapter adds nothing; erases a real self-report

Experiment	Question	What we did	Result	Verdict
E1b: coupling training	Q2: does training a model to explain its own behaviour make the explanations follow new habits?	Reproduced the paper's Hint-MMLU training on Qwen3-8B, then added the desk habit and the cheat	The paper's effect reproduced: explanations track the model's own behaviour (+9.9 points; 69% vs the paper's 79%). The habit test said "no coupling" (−2.7), but the test couldn't have shown it: the edits barely changed the habits, and nearly every model answered "no" to everything	Check passed; habit test uninformative, stopped
E2 stage 1 (9B)	Q1: does the model know its habit?	Asked each model to forecast its own choice on a side branch, then let it act	Habit models forecast the right direction at about half the size (0.38 forecast vs 0.84 act). The cheater forecasts cheating everywhere, even where it never cheats (colleague cases: 91% forecast, 0% act). The risk forecast is just the act	Knows direction, not size or when

Experiment	Question	What we did	Result	Verdict
E2 stage 1 (14B)	Q1 at a bigger size	The same, on Qwen3-14B	The habit forecast now catches the full size (100% vs 100%), but the model says the same for "a typical AI assistant". Cheat forecasts unreadable (they flip with option order)	Accurate, not self-specific
E2 stage 2	Q1: does a forecast in view correct the act?	Put the model's forecast in context under an operator rule ("leave penalties to the team")	Nothing corrected. Telling the habit model its own forecast raised the habit from 2% to 25–32%; its measured history raised it to 42%. The same words labelled "your own forecast" beat "another model's" by +17–20. Only a plain rule reminder lowered it (2.3% → 0.6%)	Self-knowledge acts as permission

Experiment	Question	What we did	Result	Verdict
E2 stage 2b	The same, under a weaker rule with room to move	A softer rule, so the habit sits at ~90%	A number note raises the act whatever the number says (a twin told "7%" doubles, 32% → 64%). A policy note is obeyed either way. Only the rule reminder cuts it (90.7% → 40.3%)	Installation at both seeds
E5 step 1	Q7: is a violation visible inside the model before it says it?	Fitted a "this is a violation" probe while Qwen3.6-27B judged runs, then read it while the model acted	The probe reads judging well (AUC 0.95) but not acting (0.55). No rise before the violation is mentioned	Unreadable
E5 wave 2	Q7: does showing the model its own judgement stop the act?	Paused 32 violating runs just before the violation and resumed them 10 times each, three ways	Left alone 90% violate; shown its own earlier judgement 29%; shown a generic rule reminder 18%. Own − reminder +11 [−1, +24]	Any reminder works; its own adds nothing

Experiment	Question	What we did	Result	Verdict
E6: main bet	Q3: does training self-prediction improve conduct?	Trained Qwen3-14B to forecast its own actions (SP), against two controls: its own actions (BO) and a generic rule (GR). Tested on 7 ODCV scenario-variants, 10 runs each	Every trained arm halved violations (untrained 73%, SP 40%, BO 36%, GR 30%). SP was no better than the controls. Its ODCV forecast broke: "yes, I will break the rules" at P = 1.00 everywhere	Uninformative for ODCV
E3 / E4	Character training and its task gate	—	A safety classifier stopped the session twice on 10-28; nothing ran	Parked with you

Insights

1. **Models know the direction of a habit, not its size, when it happens, or that it is theirs.** At 9B the forecast catches half the habit. At 14B it catches all of it, but the model says the same about any AI assistant. The cheater forecasts cheating even where it never cheats.
2. **Self-knowledge handed back in context is followed as wording, not used to correct.** A number primes the named option whatever the number is. A policy note is obeyed in either direction. The model's own stated forecast binds its next act.
3. **"Your own" is a stronger label than "another model's".** The same content gets +17–20 points more uptake when it is labelled as the model's own forecast.
4. **Only a rule reminder corrects.** It is the one note that lowered the habit in every E2 cell, and in E5 it worked as well as the model's own judgement (18% vs 29%).
5. **Recognising a violation isn't the same state as acting on one.** The model's "this is a violation" direction when judging is not present when it acts (E5 step 1).

6. **Released introspection tools don't read our implants.** The adapter reads its own training quirks only, and it overwrites the model's real cued self-report with invented ones.
7. **Any fine-tune on the model's own outputs cut ODCV violations by about half (E6).** This includes the arms with no self-prediction, so the cut comes from the fine-tuning itself, not from self-knowledge.
8. **Self-description depends on how you ask.** The risk model says it is risky in letter choices (+89) and when asked about the task (+45), but not when asked about its "unusual characteristics" (0).

Caveats

- **Small models, narrow habits.** The habits are single trained behaviours on 8–14B models (27B in E5). They may not behave like habits that come from ordinary training.
- **E6 doesn't test the main bet on agents.** Its self-forecast on ODCV collapsed to "yes" for everything, because six ODCV "yes" labels were oversampled in training. One seed, 800 steps, 7 scenario-variants, and wide intervals (SP − GR +9.6 [−18.6, +34.3]).
- **E6's cut in violations could be a side effect.** SP also lost competence (task completion 86% vs 100%), and all three fine-tunes cut violations about equally.
- **E2 stage 2 has no frozen verdict.** The operator rule was too strong and removed the habit by itself, so the gate failed. Its descriptive pattern is clear and matches stage 2b at both seeds.
- **E5 wave 2 is inconclusive on its main contrast.** Own judgement vs generic reminder is +11 [−1, +24]; the extension to 10 runs was already used.
- **E1a used one released adapter** (the SFT-only Qwen 7-setting one). A DPO adapter or a larger base may behave differently. Its cued risk result is descriptive only: the judges agreed at κ 0.57.
- **E1b's habit test had no headroom.** The edits flipped the true answer on only 1–5 of 128 items, so "no coupling" says nothing.
- **The 14B cheat forecasts are unreadable.** They change with the order the options are shown in (34–91%).
- **The risk pair has one seed** throughout.
- **E3 and E4 never ran,** so Q4 (robustness under persona pressure) and Q5 (character training) have no data from this batch.

What it led to

The final batch (CC-F, 10-28 night to 10-29) was built from this batch's two results: only rules correct, and E6 couldn't test the main bet on agents. Its write-up is Final batch (CC-F): what we tested and found.

First-batch finding	Final-batch follow-up	What it found
Only a rule reminder corrects (E2, E5)	A0.1, R2, R5: a general value line ("As an assistant, I must not…") on ODCV agents	On Qwen it cuts a violation it never names (−23 to −30), on new scenarios and in the user turn too; on MiniMax it barely moves (−4)
A self-description installs as wording (E2 stage 2b)	A0D, R3: tell the model "you tend to {violation}"	Backfires on the desk (+26–28); on ODCV it doesn't (−9, unclear)
E6's ODCV forecast collapsed, so the main bet went untested	A1: train a self-worded value statement (V5) against an impersonal twin (V5i), then a prefill test	V5 violates less in all 4 seed pairs (−16), but the competence gate fails. Prefilled, any stated principle helps (−33 to −39); the "I" framing adds little
E1a's adapter erased the cued risk report	R4: the same test on 30 questions	Erasure confirmed (−47)

Spend and set-up notes

The first batch cost **$143.45**, booked in `character_conduct/ledger.jsonl`.

Experiment	Spent	Notes
E5 (step 1 + wave 2)	$58.40	Wave 2 $30.08: 960 resumed ODCV episodes on Modal sandboxes
E6	$33.45	Three Qwen3-14B fine-tunes on A100s; adapters in the private HF repo `zhenhong1101010/cc-e6-adapters`
E1b	$25.27	$5.11 of it lost when the 10-28 hotspot outage killed the first launch

Experiment	Spent	Notes
E1a	$16.09	GPU $8.63, judges $7.46; the 15 14B adapters are kept locally for E2
E2 (four stages)	$10.25	One A100 pod per stage; no API except the 14B external predictor ($0.49)
E3 / E4	$0	Stopped by the classifier before anything ran

Set-up lessons kept for the final batch:

- The launcher used to terminate pods after ~20 minutes without contact. After the outage it was changed to wait unless RunPod says the pod is gone.
- ODCV ran on Modal here. Modal's spend limit was hit on 10-29, and the final batch moved ODCV sandboxes to RunPod through a checked stand-in.